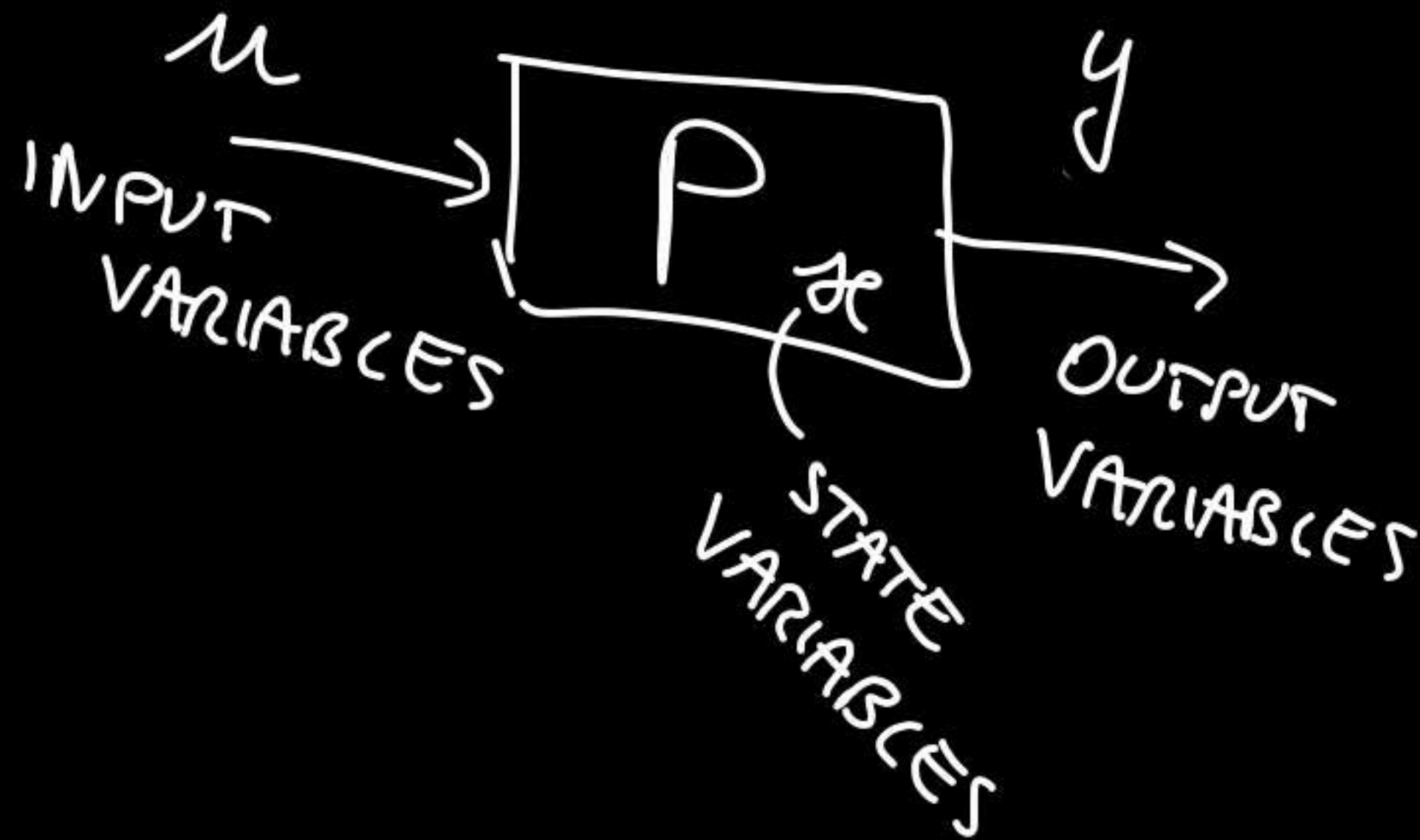


23/08/2026 - PIC - INTRODUCTION

①

PROCESS IDENTIFICATION AND CONTROL

PROCESS: ANY OBJECT/SYSTEM WE WANT TO CONTR
(PLANT, SYSTEM, ...).

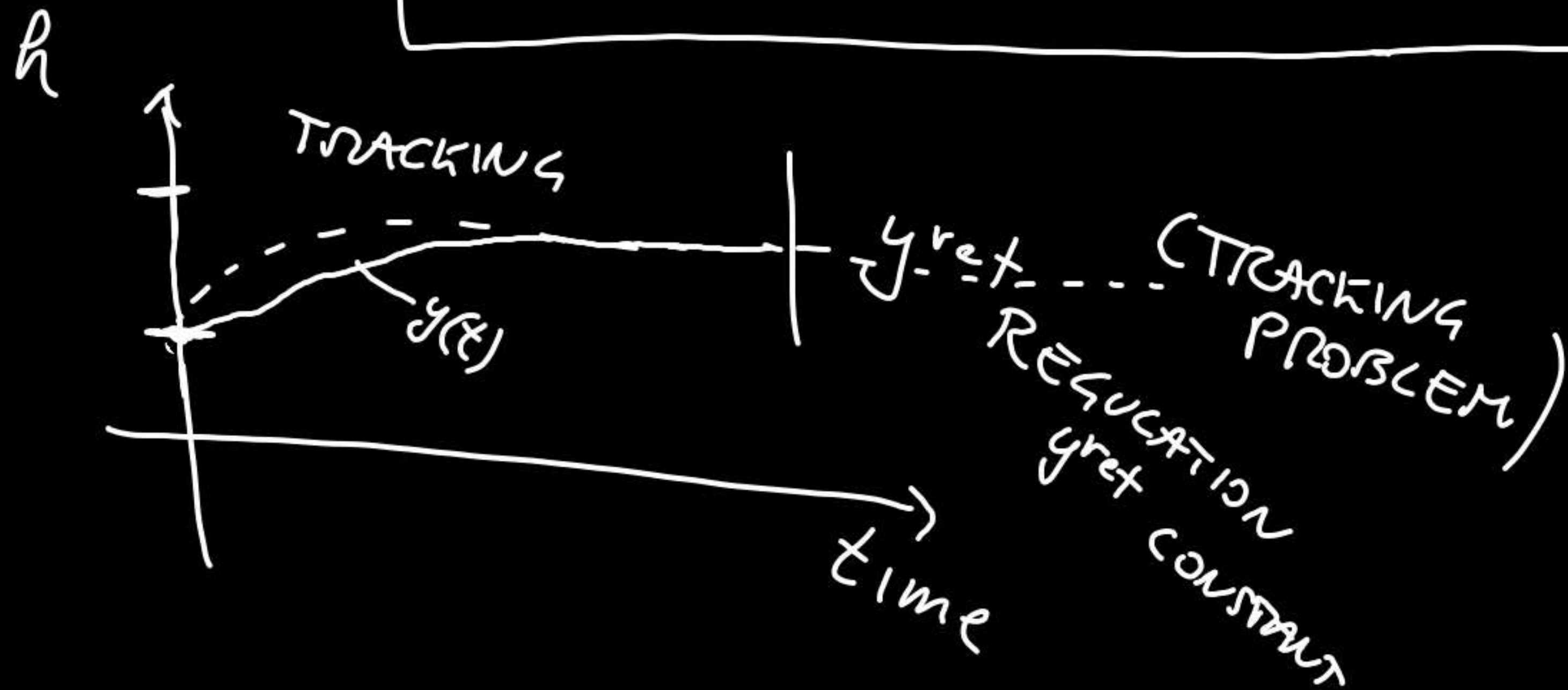
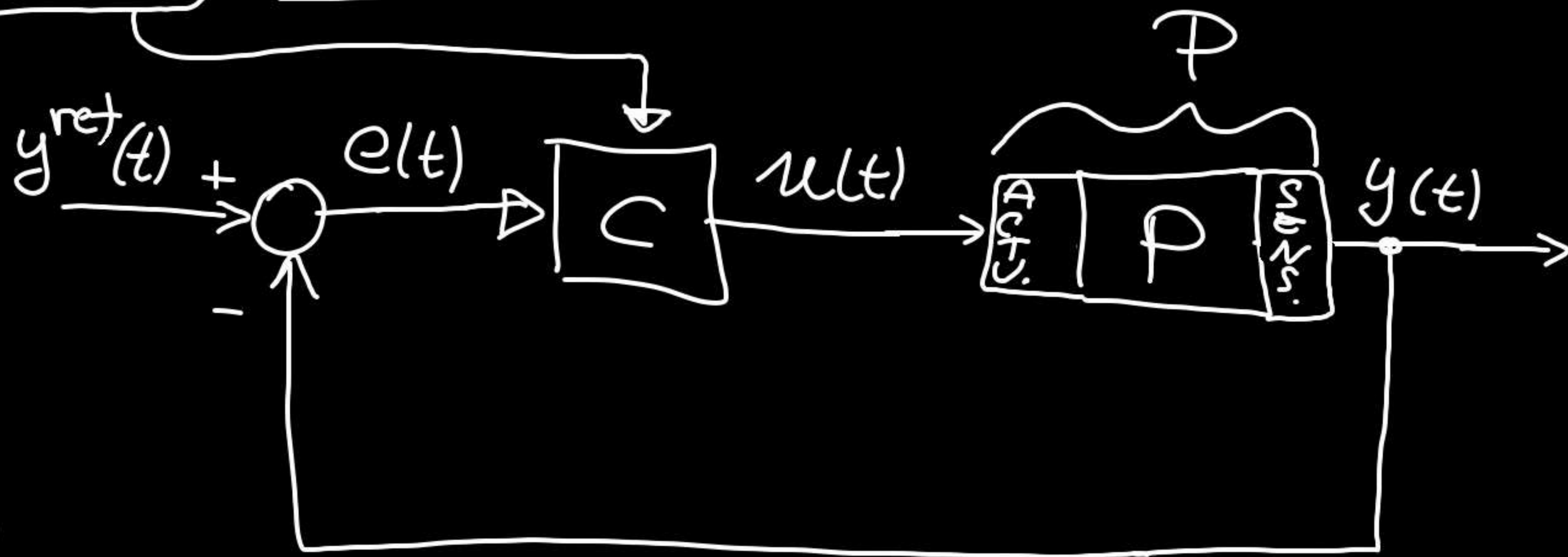


STATE SPACE
MAT. MODELS

STATE EQ.

OUTPUT EQ.

CONTROL (FEEDBACK CONTROL)

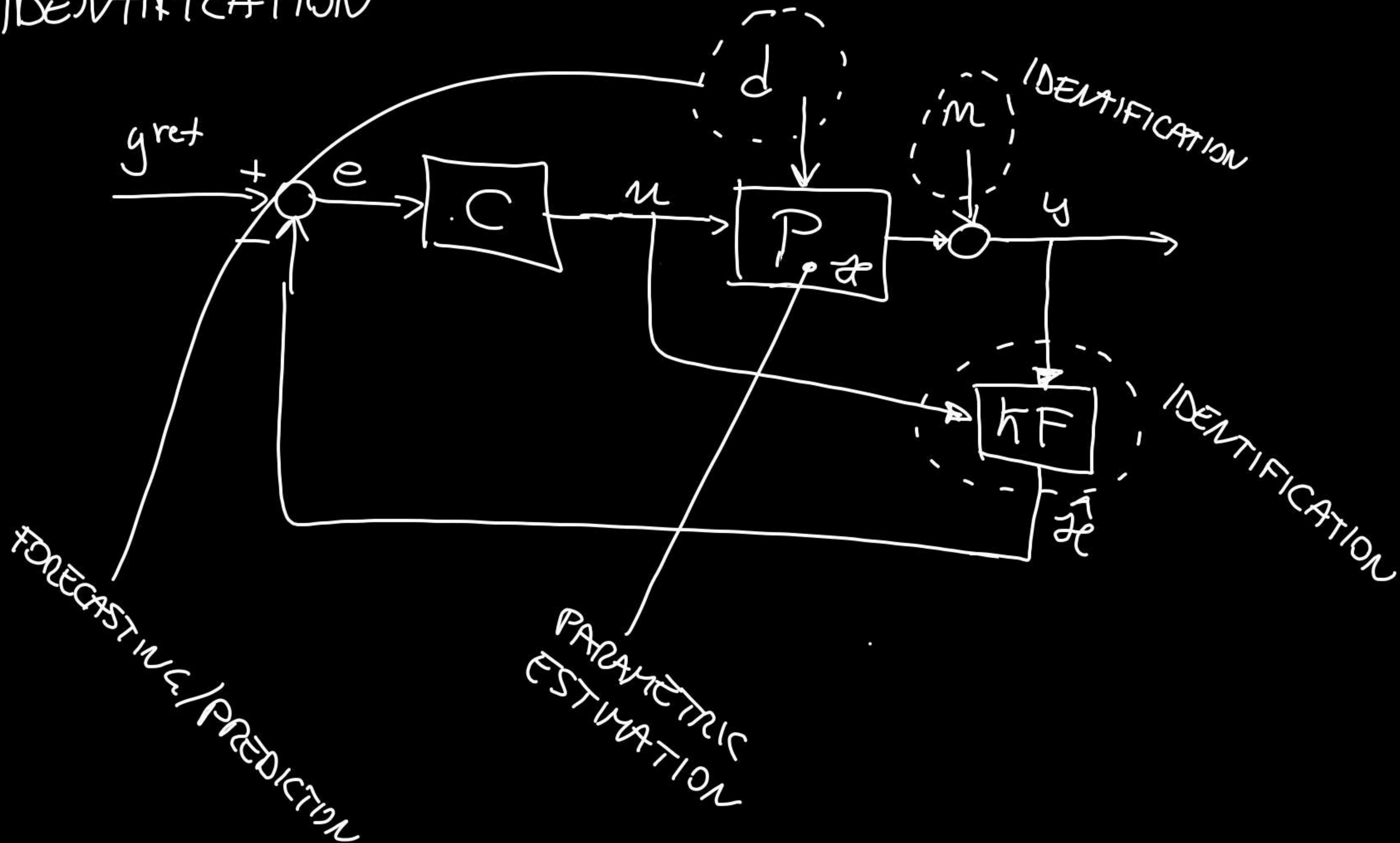


TWO CONTROL ALGORITHMS:

① MODEL PREDICTIVE CONTROL (MPC)
MODEL-BASED.

② REINFORCEMENT LEARNING (RL)
MODEL-FREE

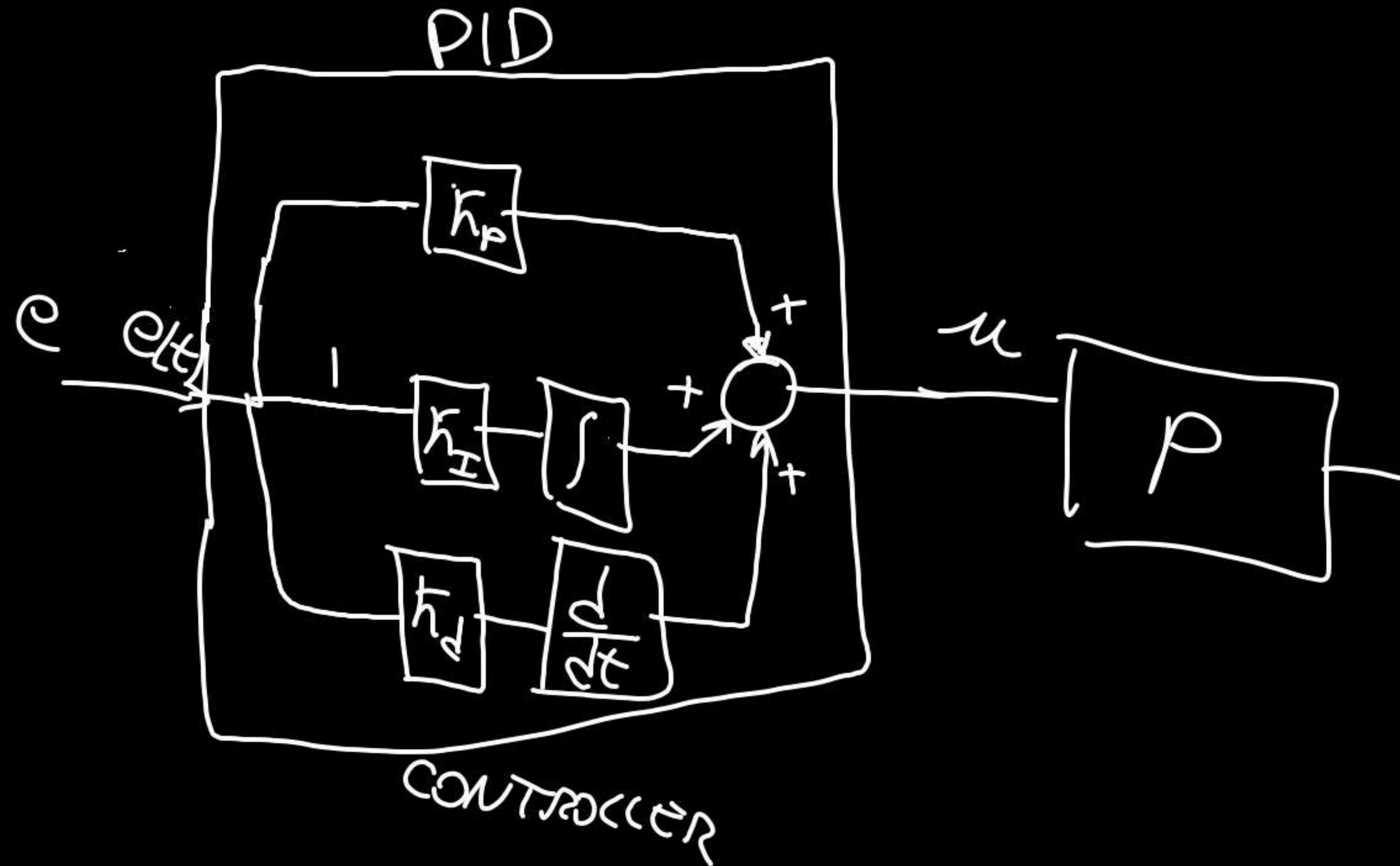
IDENTIFICATION

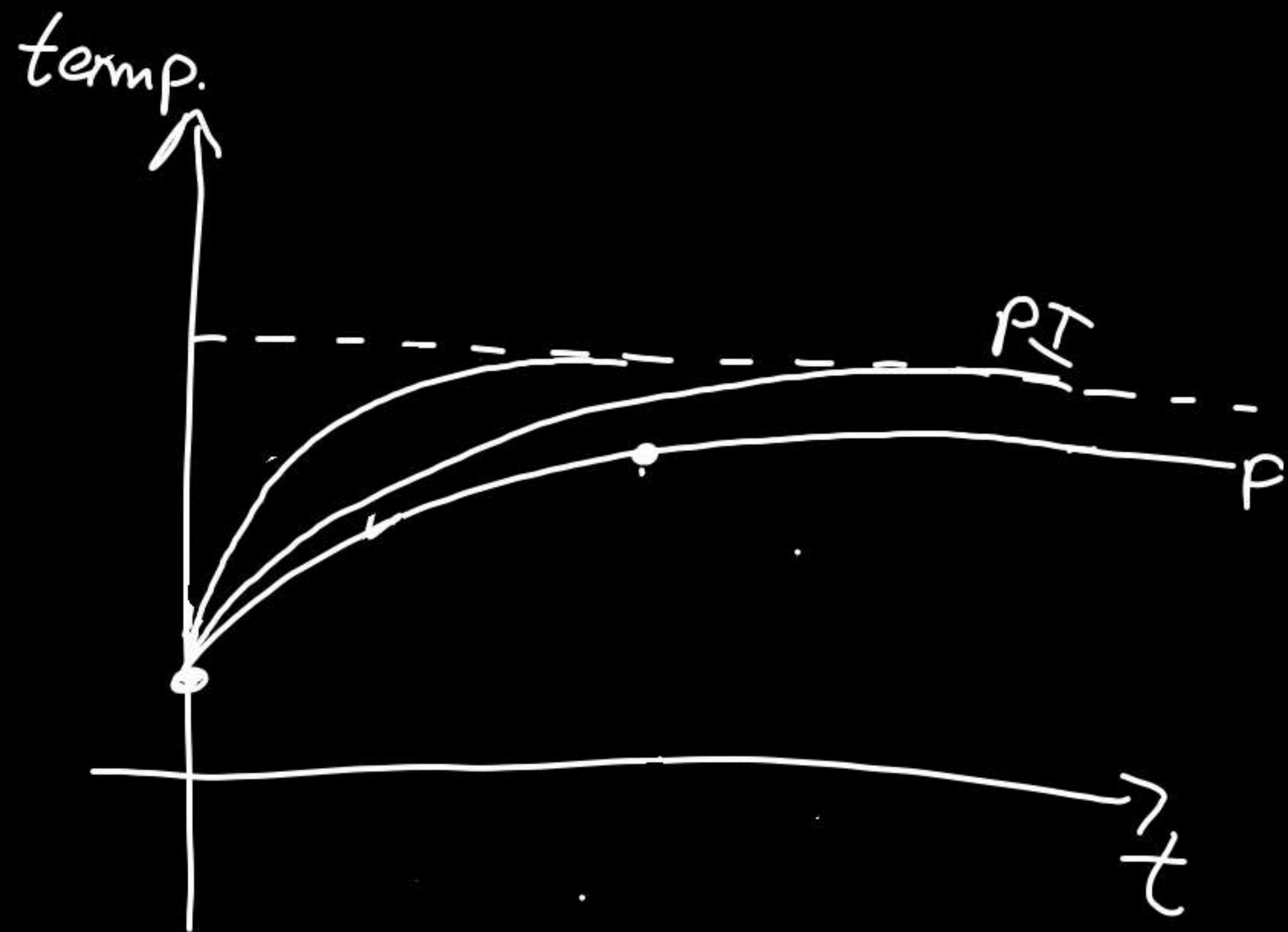


MODEL PREDICTIVE CONTROL.

• MODEL — WE NEED MODEL OF SYSTEM (STATE SPACE EQ.)

• PREDICTIVE
STANDARD WAY
(NOT-PREDICTIVE)
PID





PID TUNING: OPTIMAL SELECTION OF k_p, k_d, k_i

MODEL PREDICTIVE CONTROL

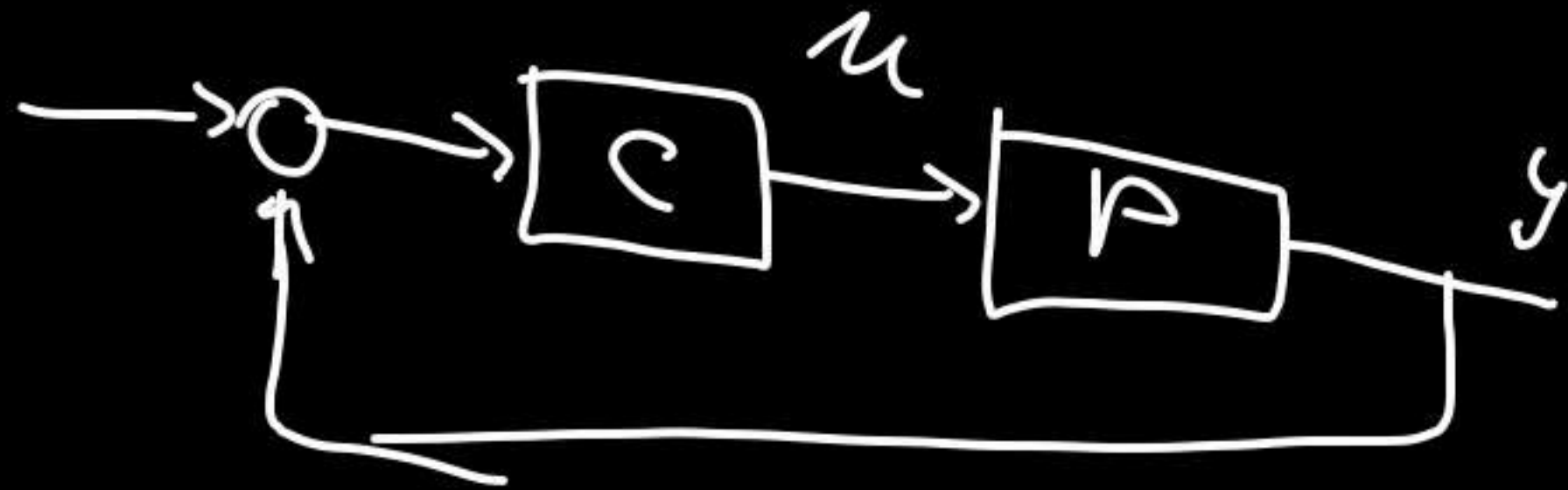
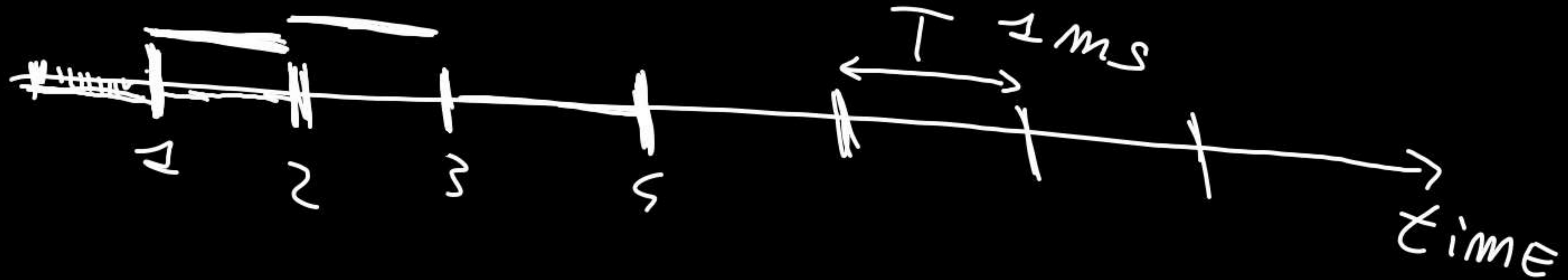
- OPTIMIZATION-BASED CONTROL TECHNIQUE.

AT ANY TIME, THE CONTROL IS FOUND BY BUILDING AND SOLVING AN OPTIMIZATION PROBLEM.

- DISCRETE-TIME CONTROL.

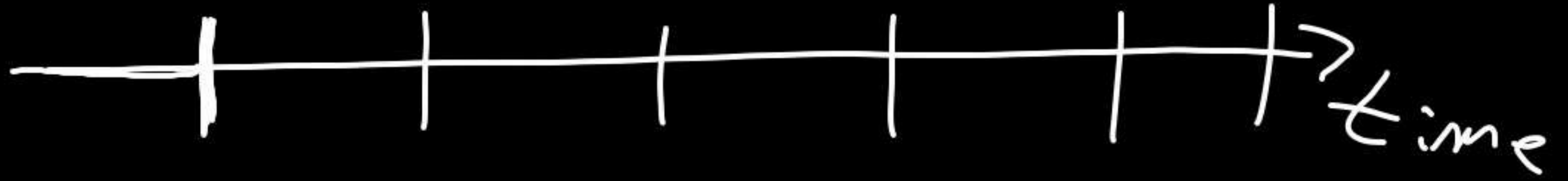
DISCRETE TIME CONTROL

- $t \in \mathbb{R}$ CONTINUOUS TIME
- $k \in \mathbb{N}$ DISCRETE TIME



MPC PSEUDOCODE

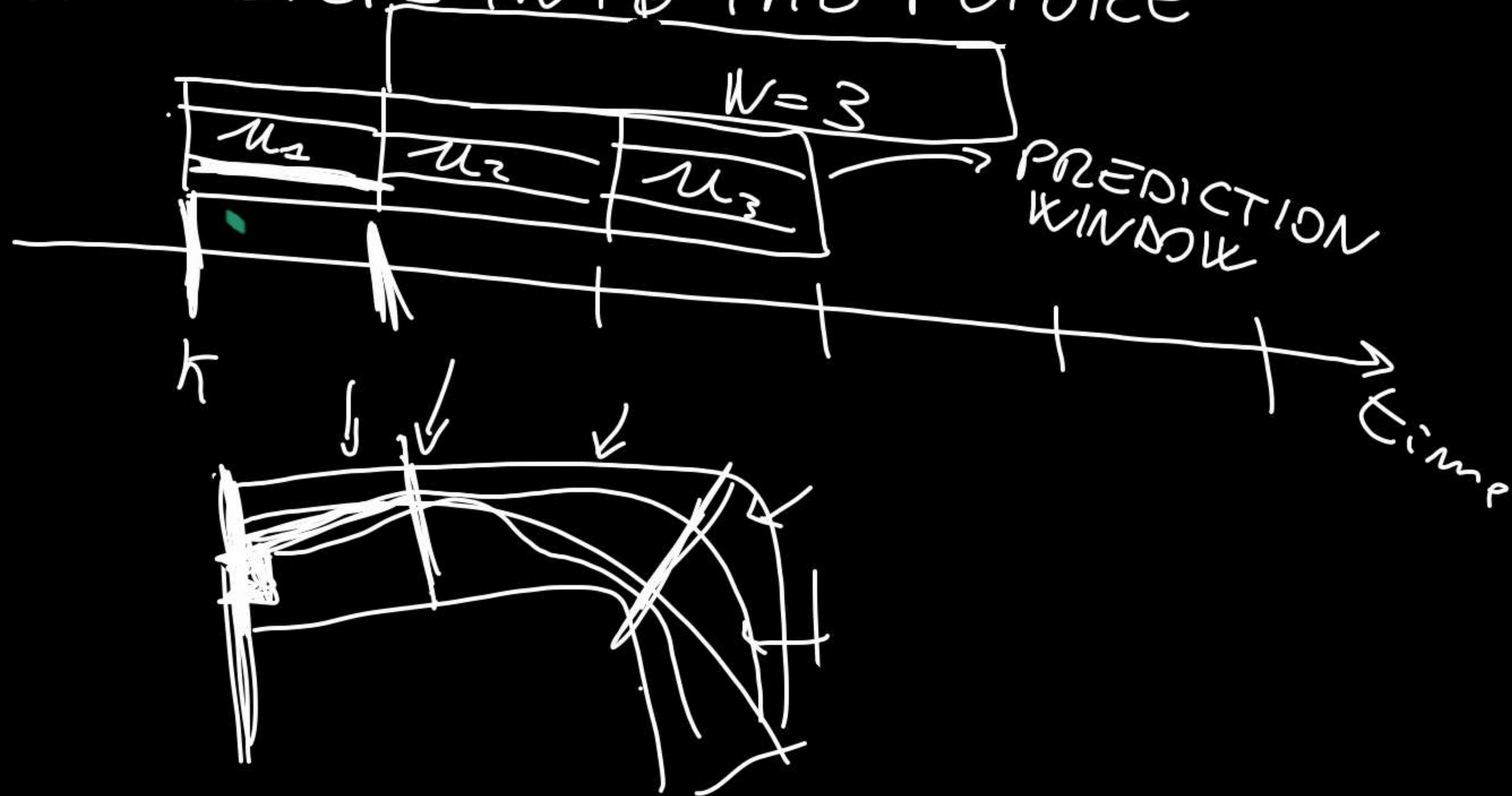
AT EVERY k :



- ① MEASURE THE CURRENT STATE OF THE SYSTEM
 $\underline{x}_k$ (ASSUME WE CAN MEASURE ALL STATE VARIABLES -
IF NOT, WE CAN ESTIMATE THEM WITH KF).
ALSO, TAKE ANY OTHER RELEVANT INFORMATION
(FOR EXAMPLE, PREDICTION OF DISTURBANCES)

② BUILD AN OPTIMIZATION PROBLEM WHOSE TIME SPAN GOES FROM NOW(k) TO $N \in \mathbb{N}$

TIME STEPS INTO THE FUTURE



③ SOLVE THE OPTIMIZATION PROBLEM (P_*)
AND WE APPLY TO THE SYSTEM THE
FIRST CONTROL ONLY (WE DISCARD THE REST).
REPEAT FROM POINT 1.

RECEIVING CONTROL.

MPC : THREE PROPERTIES.

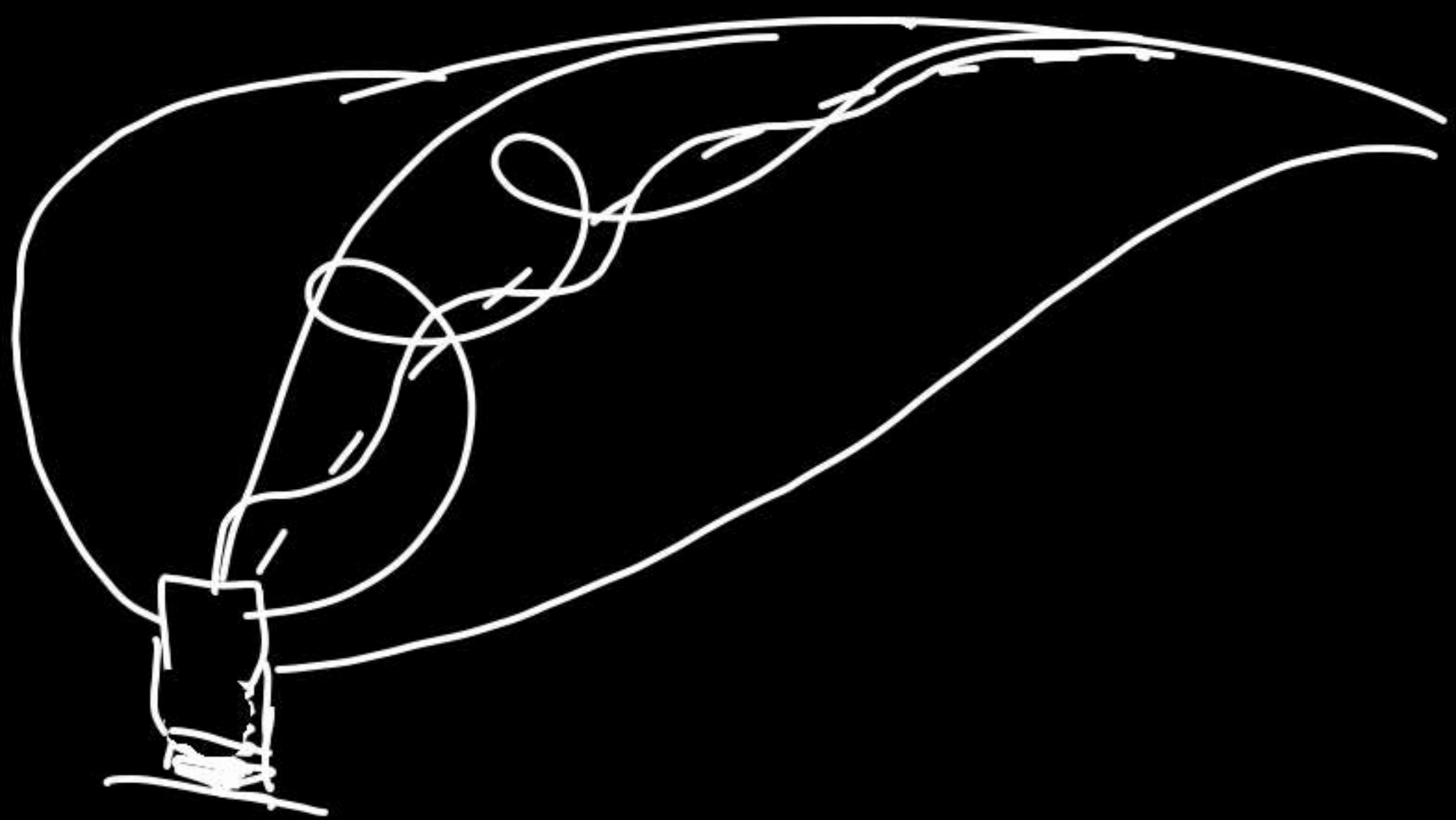
→ RECURSIVE FEASIBILITY;

MPC ALGORITHM IS RECURSIVE FEASIBLE IF,
IF THE MPC OPTIMIZATION PROBLEM IS
FEASIBLE AT ONE TIME, THEN IT WILL BE
FEASIBLE FOR ALL FUTURE TIMES.

- SAFETY.

- AVAILABILITY OF CONTROL.

— SECOND PROPERTY: OPTIMALITY.



PROBLEM: REACH ORBIT.

① STANDARD WAY (CLASSIC TRACKING/
REGULATION PROBLEM)

GOAL → TRANSLATE INTO
REFERENCES FOR
THE CONTROLLED
VARIABLES

CONTROLLER
TRACKS REF.

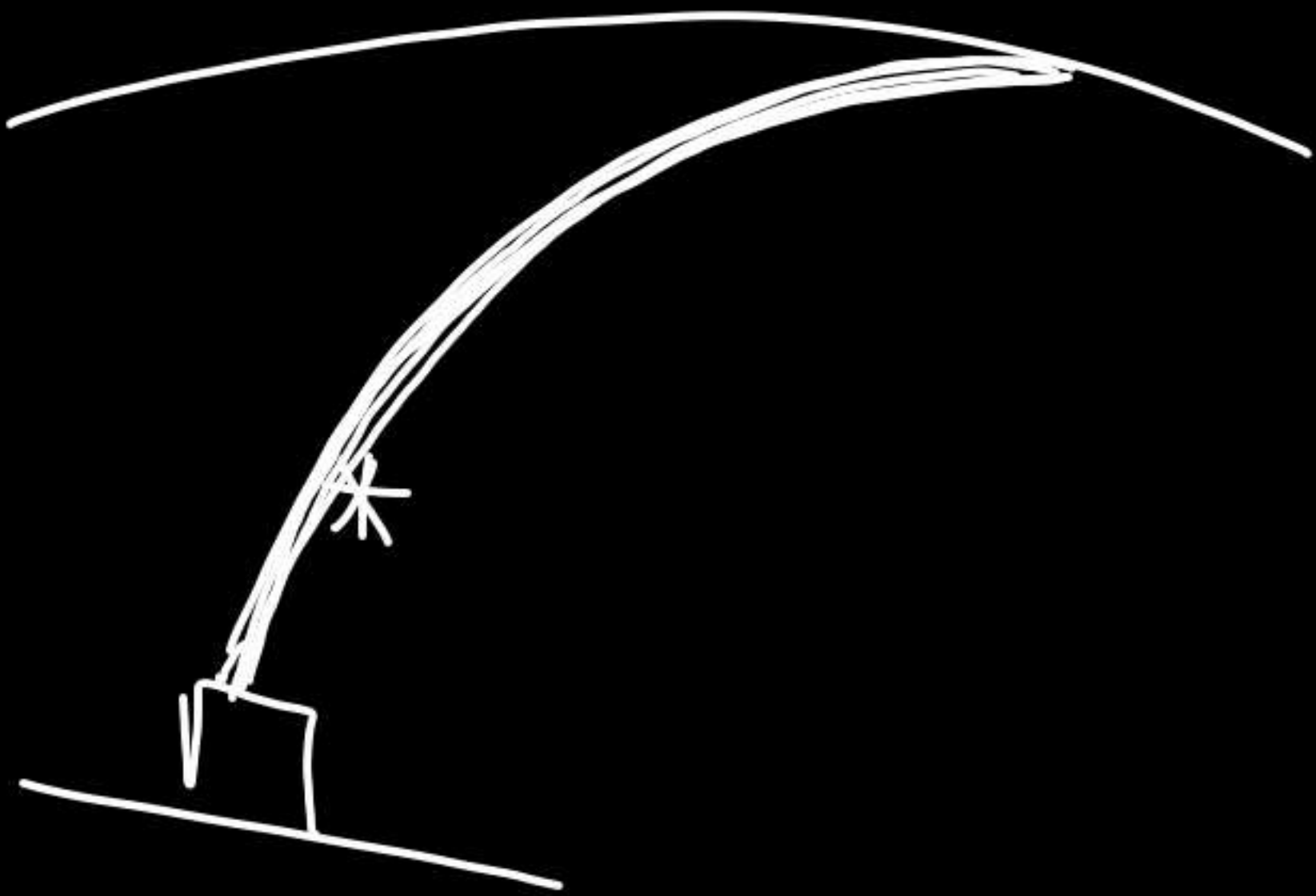
② RECENT CONTROL APPROACH: DIRECTLY OPTIMIZE THE CONTROL GOAL.

① TRACKING MPC

② ECONOMIC MPC.

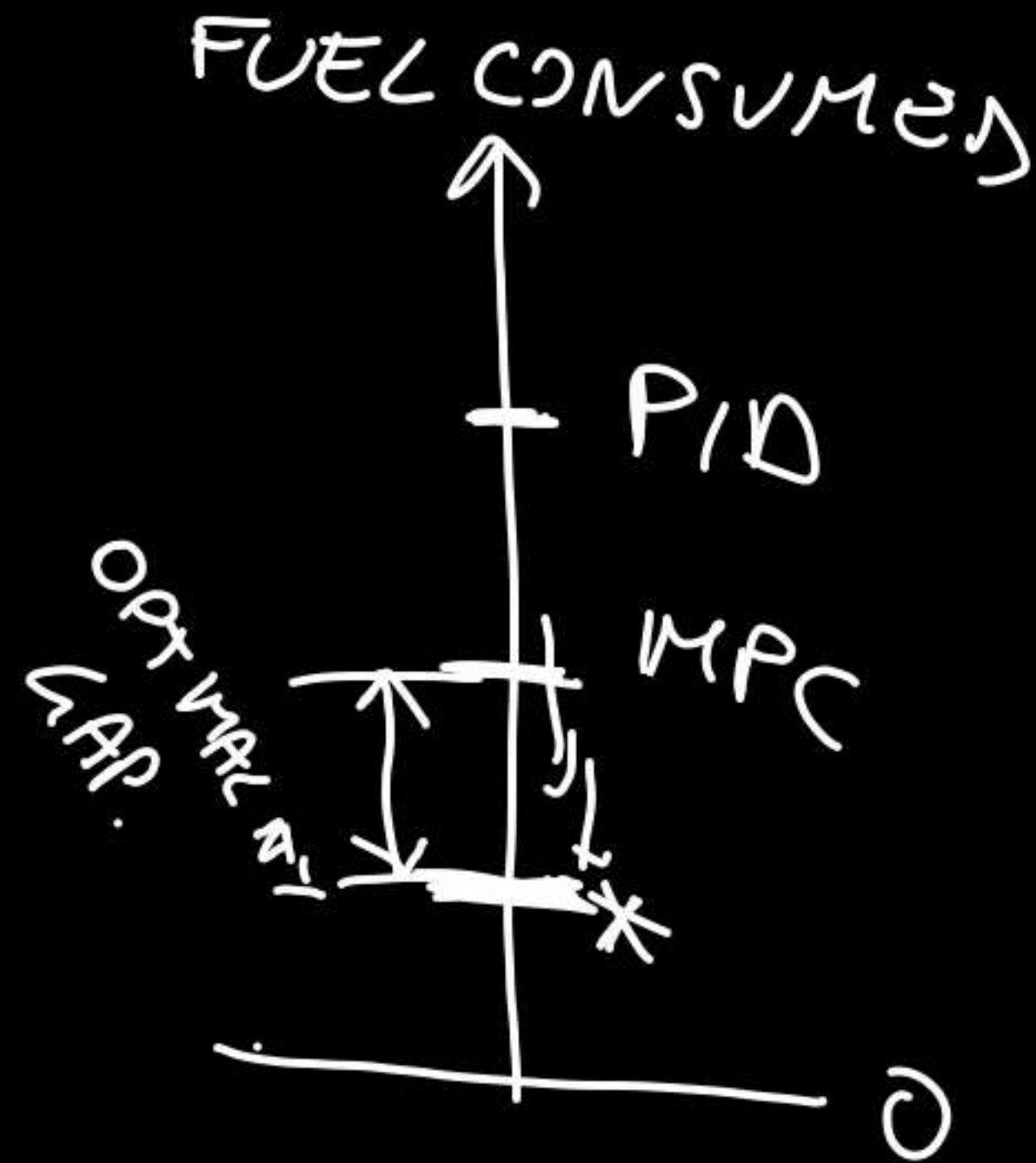
OPTIMALITY PROPERTY. (PERFORMANCE)

FOR MOST OF THE PRACTICAL PROBLEMS, THERE IS ONE (OR MANY) OPTIMAL SOLUTION(S)



REMARK:

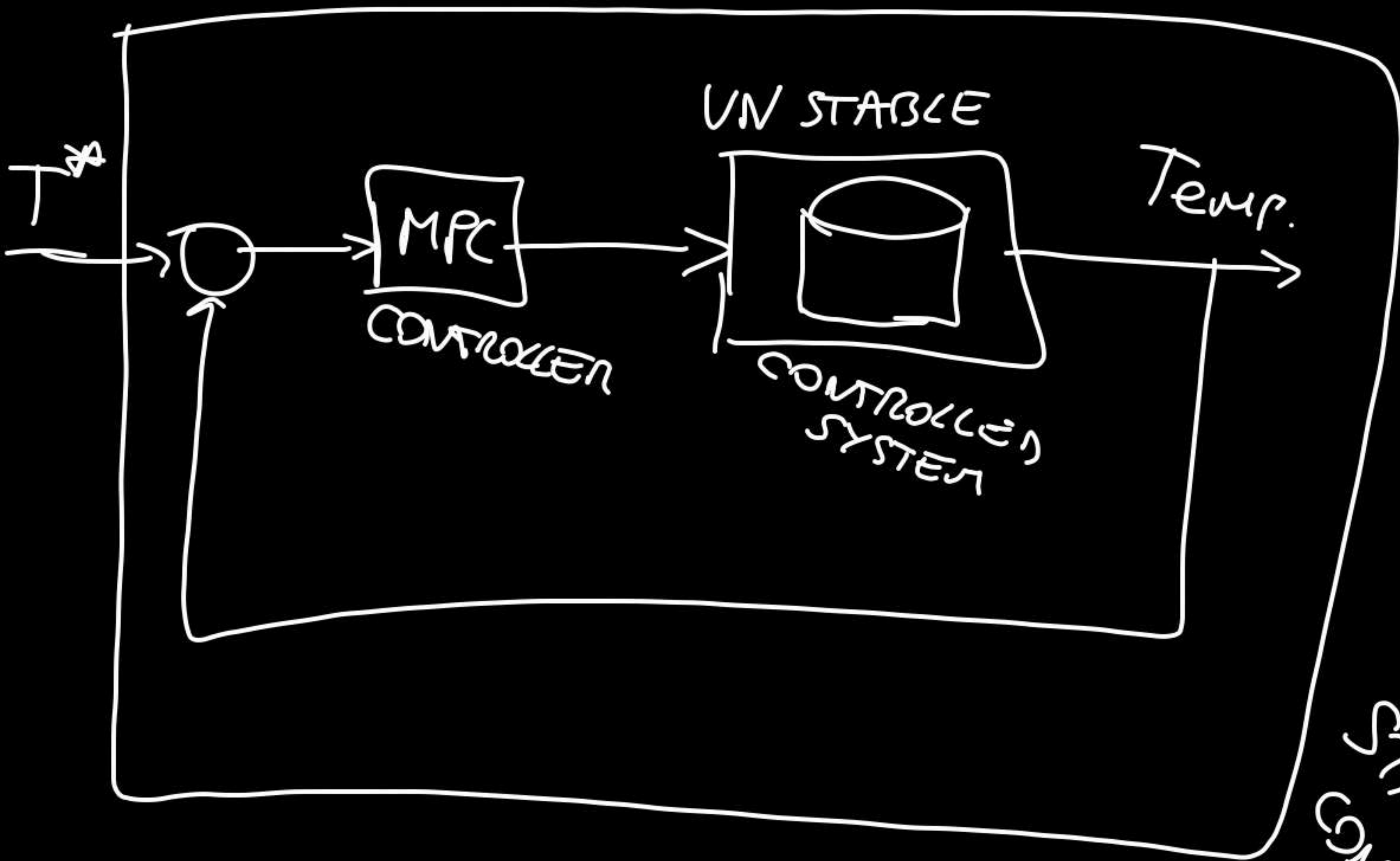
IN SOME CASES, MPC CAN BE BUILT TO HAVE OPTIMAL PERFORMANCE.



③ STABILITY PROPERTY. (SAFETY)

- SOME CHEMICAL REACTIONS.
 - MECH. SYSTEMS (PLANES, ROBOTS)
- } ARE NOT STABLE

UNSTABILITY $\longleftrightarrow$ VARIABLES DIVERGE FROM
REFERENCE.

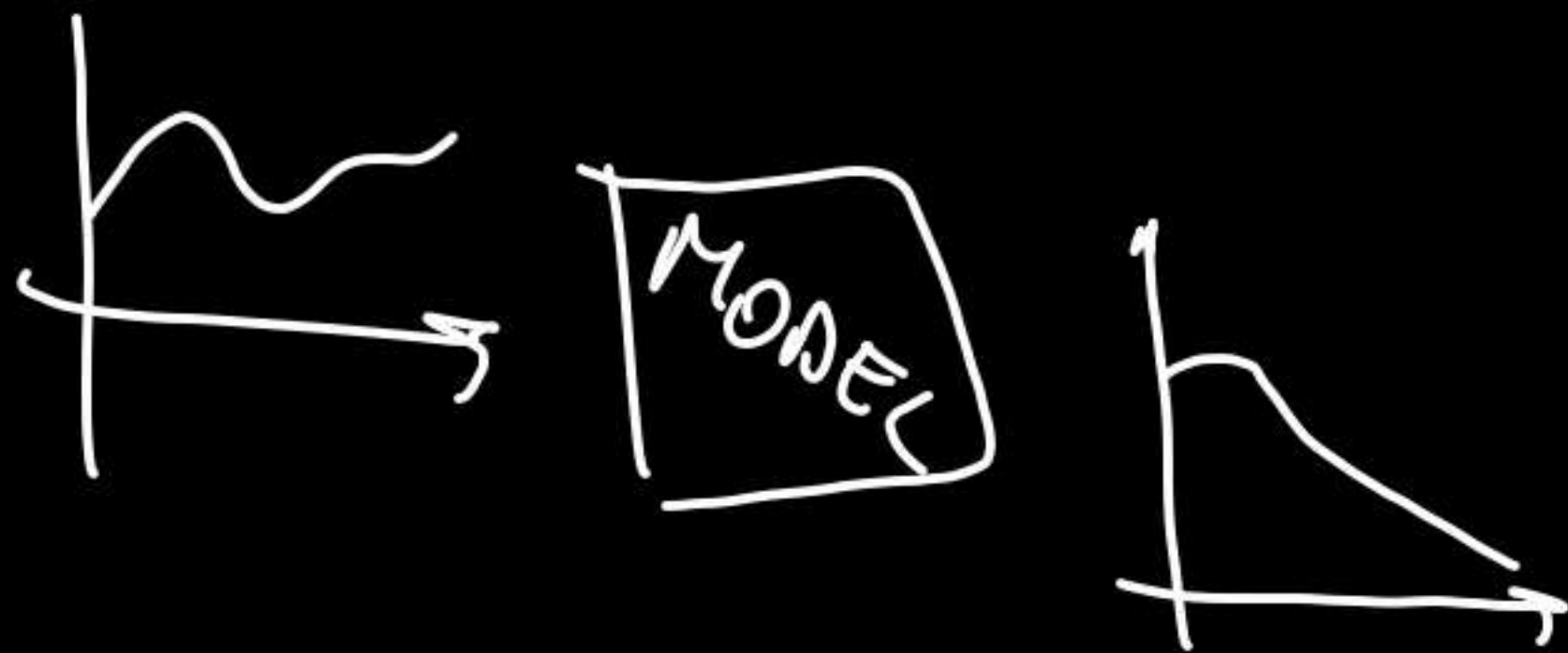
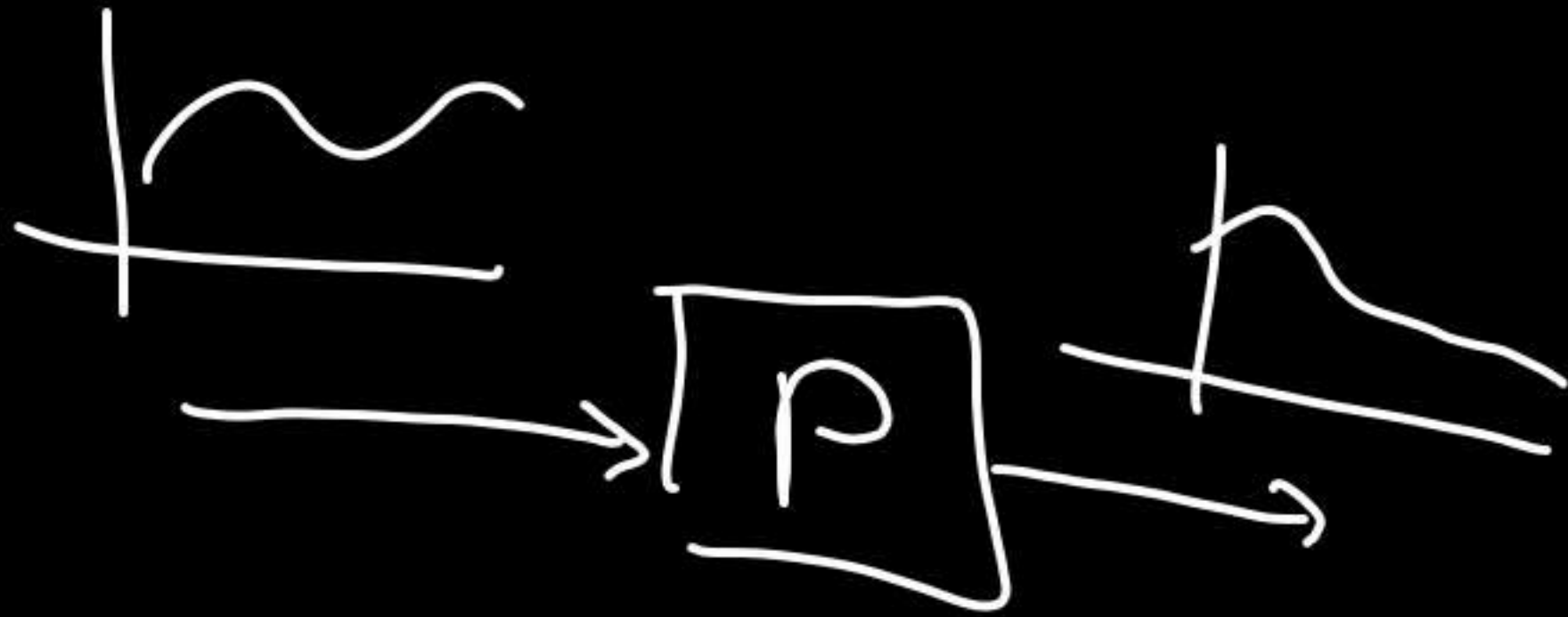


STABLE
CONTROL SYSTEM

NEW SYSTEM!

NOMINAL MPC - WE ASSUME TO HAVE A PERFECT MODEL
OF THE SYSTEM.

IT SIMPLIFIES THE STUDY OF THE THREE PROPERTIES.



ROBUST MPC. IT DEALS WITH UNCERTAINTIES IN
THE MODEL, IN THE PREDICTIONS, ETC.

MPC OPTIMIZATION PROBLEM:

① (LINEAR) CONTINUOUS PROGRAMMING. THE PROBLEM
VARIABLES ARE CONTINUOUS. SIMPLE TO SOLVE.

② MIXED-INTEGER PROGRAMMING (MIP).

- SOME VARIABLES ARE REAL. ANY VARIABLE LINKED WITH PHYSICS.

- INTEGER VARIABLES (BOOLEAN VARIABLES)

$\{0, 1\}$ • USED TO MODEL LOGICAL CONSTRAINTS,

DECISIONS TO TAKE

↓
IF → THEN

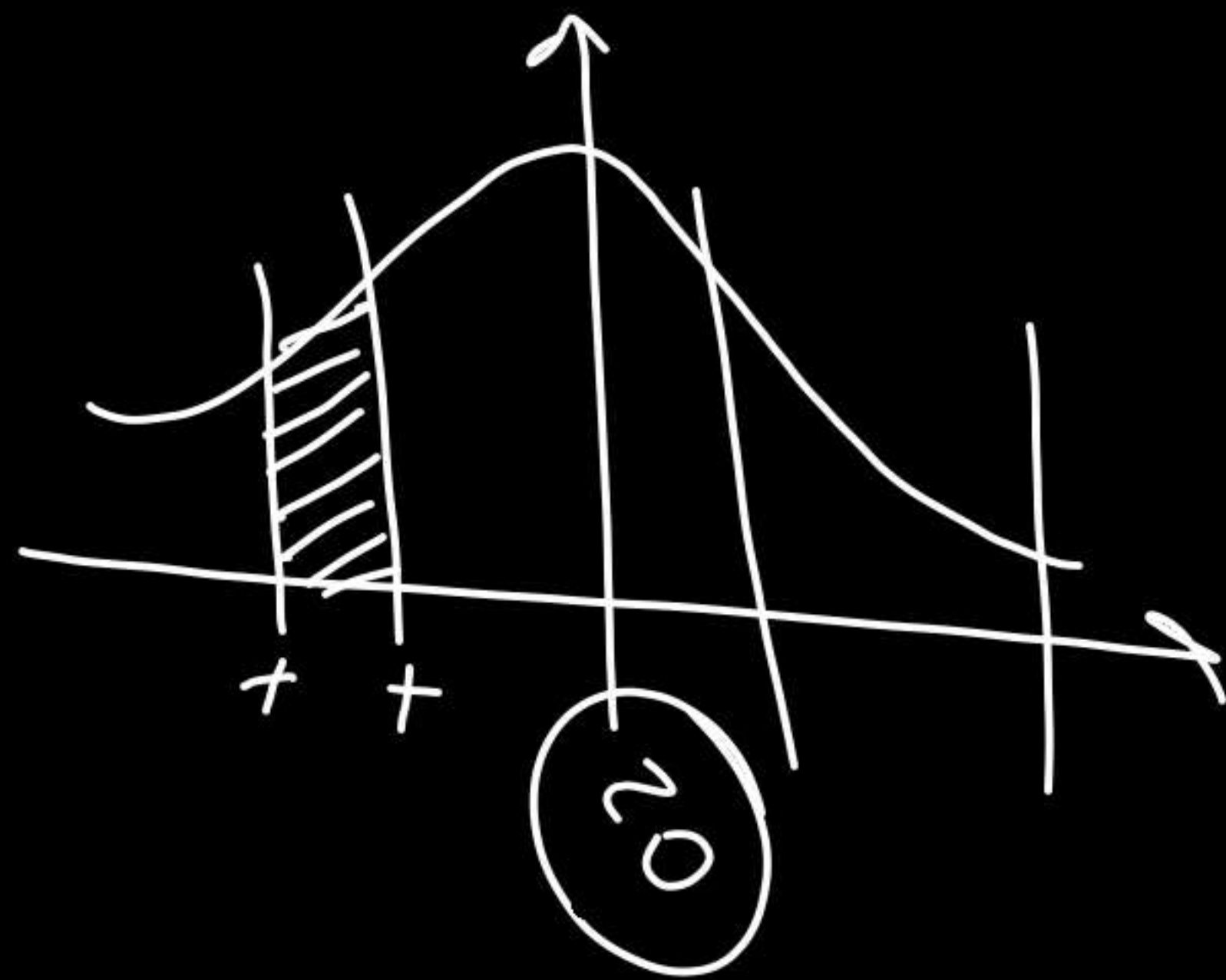
→ MIP (LINEAR) → STILL DOABLE

→ NON-LINEAR INTEGER PROGRAMMING → DIFFICULT!

23/08/2024

MOST OF PRACTICAL MPC PROBLEMS ARE MIXED-INTEGER

PROBABILISTIC MPC :



$$\max E(R) - \sigma(R)$$

$$t_i \leq 100^\circ - t_i$$
$$P(t_i \geq 100) \leq \varepsilon$$